SCONE: SKETCH-GUIDED SPATIAL GATING FOR MULTI-OBJECT 3D SCENE RECONSTRUCTION

ABSTRACT

We present SCONE (Sketch-Conditioned Object NEtwork for 3D Scene Generation), a sketch-conditioned framework for multi-object 3D scene reconstruction from a single RGB image. Traditional RGB-based methods struggle to preserve object boundaries and geometric structures, especially in low-texture or cluttered scenes. To address this, we innovatively introduce geometry-rich sketch representations into complex multi-object scene generation to enhance object-level geometric details. To further strengthen low-level geometric constraints, we propose a Spatial Gating Attention (SGA) mechanism that fuses sketch-derived structural cues with DiT latent features by using sketch geometry as an explicit gating signal. SCONE requires no additional data annotations and enables end-to-end training by incorporating auxiliary segmentation and sketch generation modules. Experiments demonstrate that SCONE significantly improves geometric consistency, boundary sharpness, and multi-object stability, establishing a simple yet effective paradigm for structure-enhanced 3D scene reconstruction.

***Index Terms*—** Sketch-based 3D generation, Spatial Gating Attention,Multi-object reconstruction

1. INTRODUCTION

The growing demand for immersive 3D content in virtual and augmented reality has accelerated progress in 3D generation, where the goal is to synthesize coherent 3D assets from compact inputs. Diffusion models and Transformer-based architectures have improved object-level fidelity and multi-view consistency [1,2]. However, extending these methods to complex multi-object scenes remains difficult, as it requires inferring spatial layout, depth ordering, and inter-object relations from limited observations. Single-image 3D scene generation must reconstruct complete environments from one RGB image [3,4], but RGB-only approaches are sensitive to occlusion and clutter, often producing distorted boundaries and inconsistent geometry.

Human relies on structural cues such as contours and edges to interpret scenes. Sketches naturally encode silhouettes and geometric information complementary to RGB appearance. Sketch-conditioned generative models have shown improved structural controllability [5], but effectively integrating sparse sketch signals into diffusion-based multi-object scene reconstruction remains challenging.

We propose **SCONE (Sketch-Conditioned Object NEtwork for 3D Scene Generation)**, the first **end-to-end sketch-conditioned framework** for single-view multi-object 3D scene reconstruction. SCONE injects object-level geometric priors by combining sketch generation and object segmentation within the diffusion pipeline, without additional annotation. Instead of treating sketches as uniform auxiliary inputs, we use them to guide structure-critical regions through a novel **Spatial Gating Attention (SGA)** module embedded in DiT layers. SGA uses sketch-derived gating maps to modulate patch-level attention, emphasizing boundaries and silhouettes while preserving generative flexibility.

By coupling sketch generation with object-level segmentation, SCONE produces instance-wise sketches that suppress background noise and provide fine-grained structural cues, improving spatial disentanglement and reducing inter-object interference. Experiments show consistent gains in geometric consistency, boundary sharpness, and multi-object stability with minimal architectural overhead.

Our main contributions are threefold:

- We propose SCONE, the first end-to-end sketch-conditioned framework for single-image multi-object 3D scene reconstruction with diffusion models.
- We introduce Spatial Gating Attention (SGA), which modulates DiT features using instance-level sketch-derived gating maps for structure-aware attention.
- We demonstrate through extensive experiments and ablations that SCONE substantially improves geometric fidelity, boundary accuracy, and scene-level stability.

2. RELATED WORK

Single-view 3D scene reconstruction has gained increasing attention in robotics, AR/VR, and digital content creation. Recent methods based on NeRF, diffusion transformers, and latent 3D generative models have improved scene-level geometry inference from a single RGB image. However, RGB-only inputs remain inherently ambiguous, particularly for low-texture regions and cluttered indoor scenes. Approaches such as PixelNeRF [6], DUSt3R [7], Make-It-3D [8], and

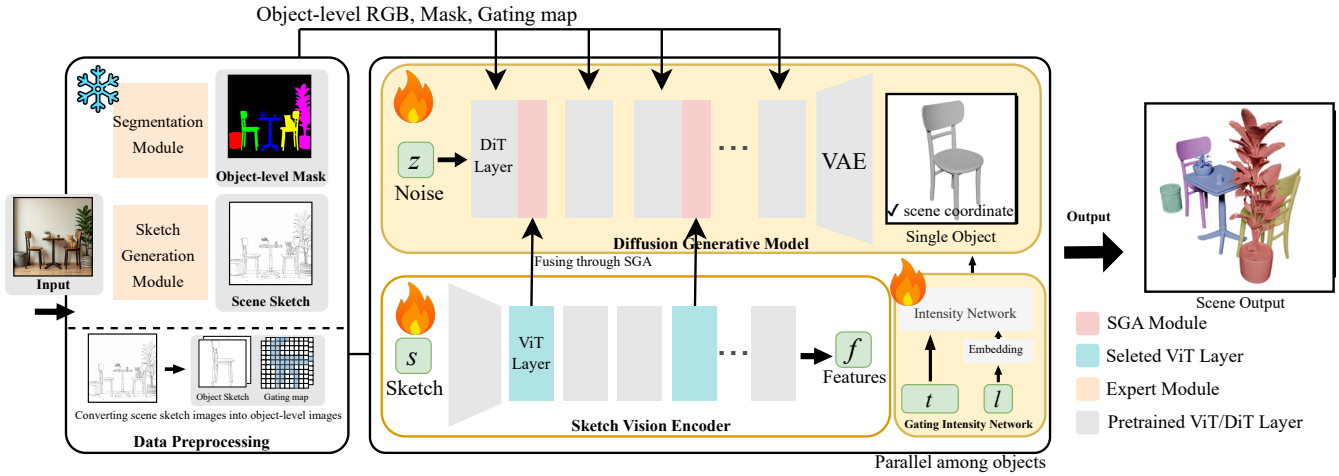


Fig. 1: Overview of our method. SCONE requires only one RGB image and reconstructs the scene with high precision.

MIDI-3D [3] achieve strong reconstruction performance but still struggle with fine object boundaries and detailed layouts.

Sketches provide complementary structural cues that are difficult to recover from RGB alone. Prior work shows their effectiveness in image generation [9, 10], shape retrieval [8] and 3D generation [11]. Sketches encode contours and boundaries that are critical for geometric reasoning. Motivated by this, we incorporate sketches as additional modalities to enhance structure-awareness in scene reconstruction.

To fuse RGB appearance with sketch-based structure, we employ a spatial gating mechanism. Gated attention and spatial gating have been used in multimodal learning and efficient vision transformers to emphasize informative regions while suppressing noise [12]. Our Spatial Gating Attention module aligns sketch features with image latents at the patch level to provide fine-grained structural control.

Overall, our framework unifies single-view 3D reconstruction, sketch-guided structural modeling, and spatial gating attention to improve geometric consistency and boundary accuracy in 3D scene reconstruction.

3. METHOD

In this section, we first describe the overall architecture of SCONE, and then introduce the sketch generation approach and Spatial Gating Attention in detail.

3.1. Overall Structure

As shown in Fig. 1, SCONE is a diffusion model for 3D scene generation, consisting of a main diffusion model and auxiliary models for sketch and mask generation. Its expert models include: **Segmentation model**, **Sketch generation model**, and **Sketch vision encoder**.

Like typical 3D diffusion generation models, at iteration i, the diffusion model takes a latent z_i and denoises it through multiple DiT layers to produce z_{i+1}. In each DiT block, RGB information from the input image is processed by a vision encoder, and its output features are used as cross-attention inputs to interact with the latent representation. Additionally, SCONE introduces a new branch: after fusing RGB information, object-level geometric sketch features are integrated with the latent representation.

Unlike conventional DiT approaches that directly use the final output of the vision encoder, we employ SGA to fuse hierarchical latent features from the sketch vision encoder with corresponding layers in the diffusion model. This design is motivated by the observation that final vision encoder outputs primarily retain semantic information, while low-level geometric details tend to be lost [13]. Since preserving geometric information aligns with our sketch-guided objective, we integrate multi-level features from shallow to deep layers of the encoder into corresponding DiT layers. This provides clearer geometric guidance throughout the denoising process.

3.2. Object-level Sketch Generation

We aim to provide geometric information for generating each object in the scene through object-level sketches. At this stage, we need to generate: (a) structural sketches of each object I'_i; (b) gating maps G_i for object i that explicitly guide the attention distribution in SGA for the reconstruction of the corresponding object.

First, for the dataset we intend to train on, we employ a segmentation model denoted as $Seg(\cdot)$ to segment various objects in the original RGB image I, obtaining masks M_i for each object in the original image. Simultaneously, we directly input the original image into the sketch generation model $Ske(\cdot)$ to obtain the scene sketch I'. Utilizing the ac-

quired sketches and masks, we extract the sketch I_i' corresponding to each object by cropping the smallest bounding box that encloses the object.

Next, we derive the gating map from I_i' used to guide the attention weights in SGA. For each I_i', we divide it into patches and compute the gating map for object i using the following equation:

$$G_{i,p} = \mathbb{I}\left\{ \sum_{(x,y)\in\mathcal{P}_p} I_i'(x,y) > 0 \right\}, \tag{1}$$

where $\mathbb{I}\{\cdot\}$ denotes the indicator function, $\mathcal{P}_p$ represents the set of pixel coordinates in patch p, and $G_{i,p} = 1$ indicates that patch p contains visible sketch strokes. Intuitively, the gating map forces the model to confine its attention to regions containing geometric information.

3.3. Spatial Gating Attention for Sketch Fusing

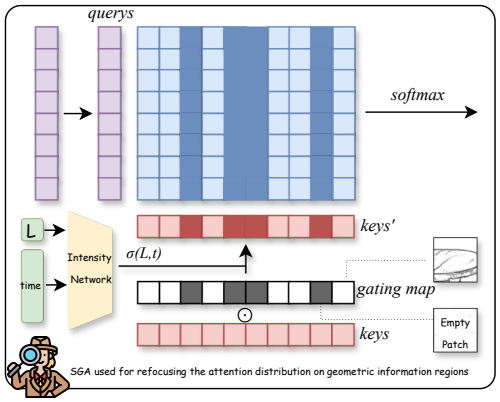


Fig. 2: The structure of Spatial Gating Attention. Intuitively, SGA redistributes the attention score to help latent reconstruct conditioning on geometry information.

To enable the model to better utilize sketch information, we design the SGA to guide the model in extracting low-level geometric features from the sketches. Given the gating map $G_i \in \mathbb{R}^{1\times HW}$, where $H \times W$ is the total number of patches in the sketch image, we aim to direct the model's attention to regions indicated by the gating map that contain geometric information. To achieve this, we introduce a learnable intensity network $k(l,t)$, where l denotes the current diffusion layer index and t represents the current denoising timestep. The attention score with explicit guidance is then formulated as:

$$A' = softmax(\frac{Q\cdot K^T}{\sqrt{d}} \odot (1 - G_i \cdot k(l,t))) \tag{2}$$

Where Q is computed from DiT latent and K is computed from sketch features. In this equation, before performing the element-wise multiplication $\odot$, the gating map G_i for object i is first broadcast to match the shape of the attention matrix. Since G_i is a $1 \times HW$ vector, it is broadcast along the row dimension.

To avoid manually computing attention matrix and leverage the Flash Attention [14] kernel in our implementation, we shift the application of the gating map to the key computation stage:

$$K' = (X \cdot W_K) \odot (1 - G_i \cdot k(l,t)) \tag{3}$$

And query vectors keep unchanged,then we can compute SGA score as standard attention:

$$\frac{Q\cdot K^T}{\sqrt{d}} \odot (1 - G_i \cdot k(l,t)) = \frac{Q(K^T \odot (1 - G_i \cdot k(l,t)))}{\sqrt{d}} \tag{4}$$

It should be noted that Eq. 4 holds if and only if G_i is a row vector. As in our settings, $G_i \in \mathbb{R}^{1\times HW}$ where each element represent whether this patch contains geometry information, so SGA can be implemented using Flash Attention kernel:

$$A' = softmax(\frac{Q\cdot K'^T}{\sqrt{d}}) \tag{5}$$

The SGA explicitly reduces the model's attention to non-informative regions. Since the patch features extracted by the vision encoder from the sketch contain both global and semantic information, even patches without line content may still contain certain information that could interfere with the model's focus on low-level geometric features during computation. Therefore, we employ gating to explicitly guide the model to reduce attention to visual features from patches.

The gating intensity $k(l,t)$ is predicted by a gating intensity network. This network takes the current transformer layer index l and the denoising timestep t as input, and predicts the intensity value for the given l and t. As illustrated in Fig. 1, the neural network consists of an embedding layer and a simple two-layer MLP. By manually initializing the critic network's parameters, we ensure that its output is 0.5 for all inputs at the beginning of training, allowing the network to gradually adjust its predictions during training.

During the forward pass, the latent representations from the previous calculation are used to compute the queries for the SGA, while the sketch features are utilized to compute the keys and values. The SGA module can be regarded as a plugin for the DiT layer. The optimization objective of SCONE is the same as that of the diffusion model.

4. EXPERIMENTS

4.1. Datasets

We train SCONE on the 3D-Front dataset [15]. For fair comparison, we adopt the same preprocessing protocol as MIDI [3], resulting in approximately 15K scenes. Evaluation is conducted on the official 3D-Front test set.

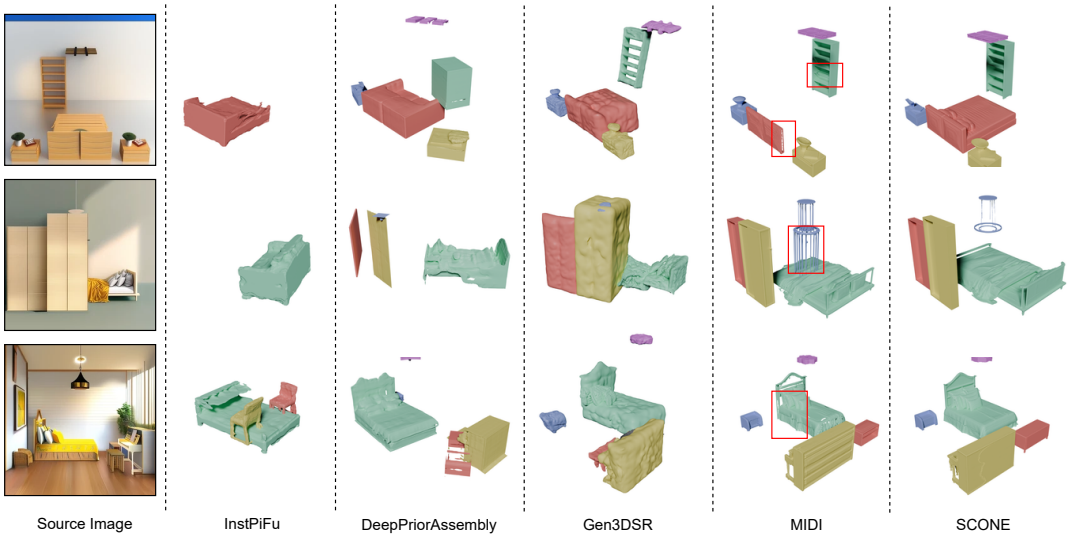


Fig. 3: Model performance display and comparison. The original image is randomly sampled from 3D-FRONT test set. By introducing strong geometric information, SCONE has achieved significant improvements in the generation details of each object compared to MIDI. The continuity, smoothness, and realism of the object surfaces have been notably enhanced.

4.2. Training Details

We employ SAM [16] as the segmentation model, Informative-Drawing [17] as the sketch generation model, and CLIP-ViT-large-14 [18] as the sketch vision encoder. For the diffusion model, we build upon the MIDI [3] foundation and continue training with our proposed modifications.

During training, both the segmentation model and sketch generation model remain frozen. The training process for the main model consists of two stages: In Stage 1, we freeze the parameters of the pre-trained diffusion model and train only the randomly initialized SGA modules and sketch vision encoder. This stage uses a learning rate of 1e-4 and trains for 4 epochs on the training dataset. In Stage 2, we unfreeze all parameters and employ LoRA [19] for fine-tuning, with LoRA adapters configured with $\alpha = 32$ and rank $r = 64$ across all layers. This stage uses a learning rate of 5e-5 and trains for 6 epochs on the training dataset. For optimization, we use the AdamW [20] optimizer with a weight decay of 0.01 and maintain a fixed batch size of 4.

4.3. Baselines

We primarily compare our method with state-of-the-art methods in single-image scene reconstruction. The compared baselines include InstPIFu [21], Gen3DSR [22], DeepPriorAssembly [23] and MIDI [3]. Note that although SCONE employs sketch images, these sketches are automatically generated by an auxiliary model and do not constitute additional manual input. This ensures a fair comparison, as all methods operate with a single RGB image as their input.

Table 1: Evaluation metrics on 3D-FRONT test dataset.

Model	Object-level			Scene-level	
	IoU-B↑	F-Score↑	CD-O↓	F-Score↑	CD-S↓
InstPIFu [21]	0.288	38.09	0.168	38.69	0.141
Gen3DSR [22]	0.363	38.11	0.157	40.07	0.123
DPA [23]	-	43.13	0.138	34.68	0.133
MIDI [3]	0.518	53.58	0.103	50.19	0.080
SCONE	**0.579**	**55.10**	**0.082**	**51.56**	**0.078**

4.4. Evaluation Metrics

We evaluate our method using *Chamfer distance (CD)* and *F-score* (with a threshold of 0.002). In general, metrics can be divided into two categories: **(a) Object-level**: We compute CS(-o), F-score and IoU-bounding box to measure the quality of reconstructed objects. We compare each object against its corresponding ground-truth mesh. For fair comparison, both predicted and ground-truth meshes are normalized first. **(b) Scene-level**: We combine all point clouds sampled from each object in the scene, computing CD(-s) and F-score to capture both object geometry and the spatial layout.

4.5. Results

As shown in Table 1, our proposed SCONE surpasses all competing models across all evaluation metrics. Compared with MIDI, which serves as the initialization for the diffusion backbone, SCONE achieves substantial improvements, demonstrating the effectiveness of our approach.

SCONE yields particularly pronounced gains over the

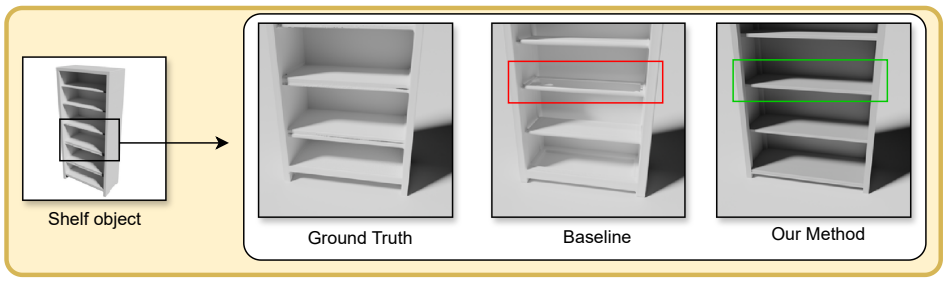


Fig. 4: Detailed comparison of object reconstruction. The baseline is MIDI [3]. The geometric information from sketches enables our method to achieve better reconstruction of object details. As shown in the figure, our method effectively resolves the object fragmentation issue present in the baseline. Our results are even smoother than the ground truth.

Table 2: Experiment results of ablation study.

Model	Object-level			Scene-level	
	IoU-B↑	F-Score↑	CD-O↓	F-Score↑	CD-S↓
Baseline	0.518	53.58	0.103	50.19	0.080
Stage 1 Only	0.530	53.40	0.116	49.29	0.129
Final Feature	0.541	53.98	0.994	51.07	0.084
Standard Attn	0.558	53.56	0.082	50.71	0.096
SCONE	**0.579**	**55.10**	**0.082**	**51.56**	**0.078**

baselines, especially over MIDI on object-level evaluations. This advantage stems from our key innovation: incorporating object-level geometric information derived from sketches. By leveraging these geometry-aware sketch cues, the model can more effectively extract the structural properties of individual objects within complex RGB scene images. Fig. 4 shows reconstruction result of different methods. Our method outperform other baselines in mesh quality and scene layout. To further explore how sketch images improves scene reconstruction, we compare the detail of reconstructed objects. Our Method effectively resolves the object fragmentation issue present in the baseline.

5. ABLATION STUDY

In this section, we conduct ablations from perspectives below: *Training Stage*, *Fusing Strategy*, and *Attention Structure*.

5.1. Ablation Study

We conduct ablation studies to validate key design choices in SCONE. First, we examine the two-stage training strategy: Stage 1 aligns old and new modalities by training only the Sketch Vision Encoder and Fusion Adapter with the base model frozen, while Stage 2 trains LoRA adapters for all modules. Table 2 shows that Stage 1 alone maintains baseline performance, whereas adding Stage 2 yields significant gains across all metrics.

Next, we compare sketch feature fusion strategies: (a) **Layer-wise fusion** injects sketch features into DiT layers at corresponding depths; (b) **Conventional fusion** only uses the encoder's final output. Strategy (a) consistently outperforms (b). This aligns with the observation that sketches convey low-level geometry—critical in early diffusion steps—while deeper encoder layers capture high-level semantics [13]. Layer-wise fusion thus provides more precise spatial guidance throughout the reverse process.

Finally, we evaluate our Sketch-Guided Attention (SGA) against standard cross-attention. Under identical training settings (Sec. 4.2), SGA surpasses standard attention on all metrics (Table 2). We attribute this to SGA's ability to constrain attention to geometrically informative regions, mitigating feature entanglement that may arise when sketch encoder latents introduce distant information into geometrically sparse areas.

6. CONCLUSION

In this paper, we propose SCONE, an innovative Diffusion-based model for 3D multi-object scene reconstruction enhanced by scene sketches. Our method addresses the challenge of declining reconstruction and layout capabilities in multi-object scenes, where image geometry and structural information are overly complex. By incorporating sketches as geometric priors, it strengthens the ability of Diffusion denoising for 3D scene reconstruction. Our approach requires no additional data annotation and automatically integrates segmentation and sketch generation models, ensuring scalability. Due to time constraints, this paper has certain limitations, such as relying on a single evaluation dataset, which restricts further demonstration of the generalizability of SCONE, which we plan to address in future work. Overall, our proposed method successfully introduces sketch information into 3D scene reconstruction, providing direction and reference for further development in the field.

7. REFERENCES

[1] Shitong Luo and Wei Hu, "Diffusion probabilistic models for 3d point cloud generation," 2021.

[2] Yichun Shi, Peng Wang, Jianglong Ye, Mai Long, Kejie Li, and Xiao Yang, "Mvdream: Multi-view diffusion for 3d generation," 2024.

[3] Zehuan Huang, Yuan-Chen Guo, Xingqiao An, Yunhan Yang, Yangguang Li, Zi-Xin Zou, Ding Liang, Xihui Liu, Yan-Pei Cao, and Lu Sheng, "Midi: Multi-instance diffusion for single image to 3d scene generation," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 23646–23657.

[4] Xiaoxiao Long, Yuan-Chen Guo, Cheng Lin, Yuan Liu, Zhiyang Dou, Lingjie Liu, Yuexin Ma, Song-Hai Zhang, Marc Habermann, Christian Theobalt, et al., "Wonder3d: Single image to 3d using cross-domain diffusion," *arXiv preprint arXiv:2310.15008*, 2023.

[5] Hail Song, Wonsik Shin, Naeun Lee, Soomin Chung, Nojun Kwak, and Woontack Woo, "S3d: Sketch-driven 3d model generation," 2025.

[6] Alex Yu, Vickie Ye, Matthew Tancik, and Angjoo Kanazawa, "pixelNeRF: Neural radiance fields from one or few images," in *CVPR*, 2021.

[7] Shuzhe Wang, Vincent Leroy, Yohann Cabon, Boris Chidlovskii, and Jerome Revaud, "Dust3r: Geometric 3d vision made easy," in *CVPR*, 2024.

[8] Junshu Tang, Tengfei Wang, Bo Zhang, Ting Zhang, Ran Yi, Lizhuang Ma, and Dong Chen, "Make-it-3d: High-fidelity 3d creation from a single image with diffusion prior," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, October 2023, pp. 22819–22829.

[9] David Ha and Douglas Eck, "A Neural Representation of Sketch Drawings," *ArXiv e-prints*, Apr. 2017.

[10] Yichen Peng, Chunqi Zhao, Haoran Xie, Tsukasa Fukusato, and Kazunori Miyata, "Sketch-guided latent diffusion model for high-fidelity face image synthesis," *IEEE Access*, vol. PP, pp. 1–1, 01 2023.

[11] Yongzhi Xu, Yonhon Ng, Yifu Wang, Inkyu Sa, Yunfei Duan, Yang Li, Pan Ji, and Hongdong Li, "Sketch2scene: Automatic generation of interactive 3d game scenes from user's casual sketches," *arXiv preprint arXiv:2408.04567*, 2024.

[12] Hanxiao Liu, Zihang Dai, David So, and Quoc V Le, "Pay attention to mlps," in *Advances in Neural Information Processing Systems*, M. Ranzato, A. Beygelzimer, Y. Dauphin, P.S. Liang, and J. Wortman Vaughan, Eds. 2021, vol. 34, pp. 9204–9215, Curran Associates, Inc.

[13] Amin Ghiasi, Hamid Kazemi, Eitan Borgnia, Steven Reich, Manli Shu, Micah Goldblum, Andrew Gordon Wilson, and Tom Goldstein, "What do vision transformers learn? a visual exploration," 2022.

[14] Tri Dao, "Flashattention-2: Faster attention with better parallelism and work partitioning," 2023.

[15] Huan Fu, Bowen Cai, Lin Gao, Lingxiao Zhang, Jiaming Wang Cao Li, Zengqi Xun, Chengyue Sun, Rongfei Jia, Binqiang Zhao, and Hao Zhang, "3d-front: 3d furnished rooms with layouts and semantics," 2021.

[16] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollár, and Ross Girshick, "Segment anything," *arXiv:2304.02643*, 2023.

[17] Caroline Chan, Fredo Durand, and Phillip Isola, "Learning to generate line drawings that convey geometry and semantics," 2022.

[18] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever, "Learning transferable visual models from natural language supervision," 2021.

[19] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen, "LoRA: Low-rank adaptation of large language models," in *International Conference on Learning Representations*, 2022.

[20] Ilya Loshchilov and Frank Hutter, "Decoupled weight decay regularization," 2019.

[21] Haolin Liu, Yujian Zheng, Guanying Chen, Shuguang Cui, and Xiaoguang Han, "Towards high-fidelity single-view holistic reconstruction of indoor scenes," in *Proceedings of European Conference on Computer Vision*, 2022.

[22] Andreea Ardelean, Mert Özer, and Bernhard Egger, "Generalizable 3d scene reconstruction via divide and conquer from a single view," in *International Conference on 3D Vision (3DV)*, 2025.

[23] Junsheng Zhou, Yu-Shen Liu, and Zhizhong Han, "Zero-shot scene reconstruction from single images with deep prior assembly," 2024.